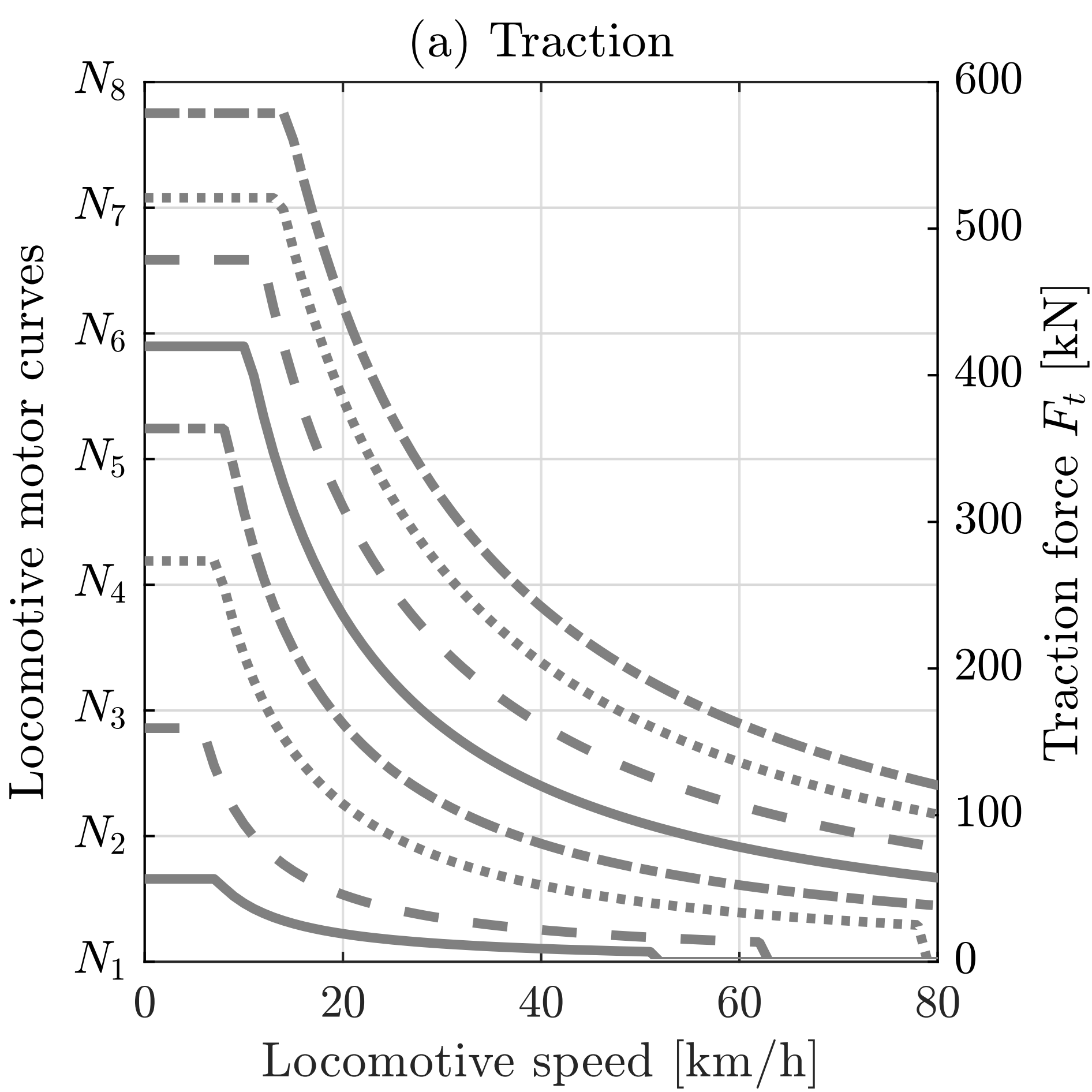

(a) Traction
N_8
N_7
N_6
N_5
N_4
N_3
N_2
N_1
Locomotive motor curves
600
500
400
300
200
100
0
Traction force F_t [kN]
0
20
40
60
80
Locomotive speed [km/h]